



IDC216 : Statistics for Scientists (2025-26)

Exercise Set

1. The room temperature (in $^{\circ}\text{C}$) at different times (in minutes) after switching on AC is given below.

Time (min)	2	4	6	8
Temperature	37	30	?	24

Estimate the temperature after 6 minutes.

2. Population count in the *Republic of Statistica* over the years is given below:

Year	2000	2005	2010	2015
Population (in thousands)	450	520	630	780

Fit a linear regression line to predict the population in 2020.

3. The following dataset shows rainfall (mm) over 10 days:

Rainfall | 12 15 18 22 25 27 28 30 32 50

Calculate the mean, median, and mode.

4. For the data in Question 3, compute skewness and comment on the shape of the distribution.

5. A dataset shows exam scores of 12 students:

Scores | 45 50 52 55 55 58 60 62 65 70 72 80

Compute mean, variance, and standard deviation.

6. A researcher records the height (cm) and weight (kg) of 6 individuals:

Height	150	155	160	165	170	175
Weight	50	52	55	58	60	65

Fit a linear regression model and predict weight for height = 168 cm. Now treat weight as independent variable and find the height for weight = 57 kg.

7. The following dataset shows growth of a plant measured over weeks:

Week	2	4	6	8	10
Height (cm)	2	4	9	16	25

Find the Lagrange polynomial for this growth and estimate the height after week 7.